

LLM 20 Questions



Baseline

Model: Gemma 7B

- Questioner + Answerer agents
- No memory / weak logic

Problems observed:

- Repetitive / generic questions
- No use of previous answers
- Weak guessing strategy
- Occasional errors in answer phase



Baseline Output

```
response="Sure, what's your first question?"
response='the empire state building'
Question 1: Sure, what's your first question?
Answer: no
Guess: the empire state building

response="Here's a question: Is the keyword a living organism?"
response='the answer to the question is: no.'
Question 2: Here's a question: Is the keyword a living organism?
Answer: no
Guess: the answer to the question is: no.

response='Sure, please ask your next question: Is the keyword a planet?'
response='mars'
Question 3: Sure, please ask your next question: Is the keyword a planet?
Answer: no
Guess: mars

response="Here's the next question: Is the keyword a mineral?"
response='the answer is: '
Question 4: Here's the next question: Is the keyword a mineral?
Answer: no
Guess: the answer is:

response='Sure, please ask your next question: Is the keyword a musical instrument?'
response='the answer is: a guitar.'
Question 5: Sure, please ask your next question: Is the keyword a musical instrument?
Answer: no
Guess: the answer is: a guitar.
```

```
response='Does the keyword have a lot of trees?'
response='forest'
Question 1: Does the keyword have a lot of trees?
Answer: no
Guess: forest

response='Does the keyword have a lot of water?'
response='the answer is:'
Question 2: Does the keyword have a lot of water?
Answer: no
Guess: the answer is:

response='Is the keyword a desert?'
response='the answer is: tokyo'
Question 3: Is the keyword a desert?
Answer: no
Guess: the answer is: tokyo

response='Does the keyword have a lot of mountains?'
response='the great wall of china'
Question 4: Does the keyword have a lot of mountains?
Answer: no
Guess: the great wall of china

response='Is the keyword a city?'
response='los angeles'
Question 5: Is the keyword a city?
Answer: no
Guess: los angeles
```

Baseline Output

Baseline Run 1

Valid guesses:
empire state building = valid
the answer to the question is: no. = invalid
mars = valid
the answer is: = invalid
the answer is: a guitar. = invalid

Win Rate
Correct guesses = 0
Total guesses = 5

Win Rate = $(0 / 5) \times 100$
Win Rate = 0%
Valid Output Rate
Valid outputs = 2
Total outputs = 5

Valid Output Rate = $(2 / 5) \times 100$
Valid Output Rate = 40%
Average Questions Used
Questions used = 5

Average Questions Used = 5 / 1 run
Average Questions Used = 5

Baseline Run 2

Valid guesses:
forest = valid
the answer is: = invalid
the answer is: tokyo = invalid
great wall of china = valid
los angeles = valid

Win Rate
Correct guesses = 0
Total guesses = 5

Win Rate = $(0 / 5) \times 100$
Win Rate = 0%
Valid Output Rate
Valid outputs = 3
Total outputs = 5

Valid Output Rate = $(3 / 5) \times 100$
Valid Output Rate = 60%
Average Questions Used
Questions used = 5

Average Questions Used = 5 / 1 run
Average Questions Used = 5

Metric	Baseline Run 1	Baseline Run 2
Win Rate	0%	0%
Valid Output Rate	40%	60%
Average Question Relevance Score	0.8 / 2.0	1.4 / 2.0
Average Context Consistency Score	0.4 / 2.0	1.0 / 2.0
Average Questions Used	5	5

Improvement v1

- Structured questioning strategy
- Uses previous Q&A context
- More logical narrowing of search space

Comparison

Aspect	Baseline	Improved
Question quality	random	structured
Logic	weak	guided
Guess accuracy	low	higher



Improved structured baseline simulation...

Question 1: Is it a place?

Answer: yes

Guess: country

Question 2: Is it a country?

Answer: yes

Guess: tanzania

Question 3: Is it located in Africa?

Answer: yes

Guess: tanzania

Question 4: Is it known for wildlife or safaris?

Answer: yes

Guess: tanzania

Question 5: Is it Tanzania?

Answer: yes

Guess: tanzania

Improvement v2

Prompt Engineering
- Few-shot Learning

Comparison

Baseline	Improved
random questions	strategic questions
hallucinated guesses	context-aware guesses
weak narrowing	progressive narrowing
generic prompting	guided prompting

```
response= Question: Is the hidden keyword a large island nation known for its historical significance and diverse culture?
response='answer:'
Question 1: Question: Is the hidden keyword a large island nation known for its historical significance and diverse culture?
Answer: no
Guess: answer:

response="Sure, here's your question: Is the hidden keyword a city that has a strong maritime presence and is known for its canals and
bridges?"
response='question:'
Question 2: Sure, here's your question: Is the hidden keyword a city that has a strong maritime presence and is known for its canals an
d bridges?
Answer: no
Guess: question:

response="Okay, here's your question: Is the hidden keyword a historical city that's been a hub for trade and commerce throughout the c
enturies?"
response='sure, guess the keyword:'
Question 3: Okay, here's your question: Is the hidden keyword a historical city that's been a hub for trade and commerce throughout the
centuries?
Answer: no
Guess: sure, guess the keyword:

response="The answer is: Is the hidden keyword a historical city that's been a major center for the spice trade?"
response='venice'
Question 4: The answer is: Is the hidden keyword a historical city that's been a major center for the spice trade?
Answer: no
Guess: venice

response='Yes, the answer is: Is the hidden keyword a coastal city known for its aromatic spices and historical significance?'
response='kolkata'
Question 5: Yes, the answer is: Is the hidden keyword a coastal city known for its aromatic spices and historical significance?
Answer: no
```



Improvement v3

LLM pipeline improvement

Comparison

Before	After
random questions	strategic questions
no guidance	guided reasoning

```
response='Is the keyword a city or a town?'
response='the answer is:'
Question 1: Is the keyword a city or a town?
Answer: no
Guess: the answer is:

response='Are the majority of the inhabitants of the place known to be tall or short?'
response='tanzania'
Question 2: Are the majority of the inhabitants of the place known to be tall or short?
Answer: yes
Guess: tanzania

response='Are the historical buildings in the place mostly made of stone or wooden materials?'
response='tanzania'
Question 3: Are the historical buildings in the place mostly made of stone or wooden materials?
Answer: yes
Guess: tanzania

response='Are the streets of the place mostly straight or winding?'
response='tanzania'
Question 4: Are the streets of the place mostly straight or winding?
Answer: yes
Guess: tanzania

response='Are the rivers that flow through the place mostly perennial or seasonal?'
response='tanzania'
Question 5: Are the rivers that flow through the place mostly perennial or seasonal?
Answer: yes
Guess: tanzania
```

